# SIMULATOR
# EXAMINATION SCENARIO GUIDE

**SCENARIO TITLE:** 14--01 NRC ESG-1

**SCENARIO NUMBER:** 14--01 NRC ESG-1

**EFFECTIVE DATE:** See Approval Dates

**EXPECTED DURATION:** 80 minutes

**REVISION NUMBER:** 00

**PROGRAM:**

| | |
|---|---|
| | **L.O. REQUAL** |
| X | **INITIAL LICENSE** |
| | **STA** |
| | **OTHER** ________________ |

Revision Summary
New Issue for 14-01 ILOT NRC exam

| | | |
|---|---|---|
| **PREPARED BY:** | G Gauding<br>**Lead Regulatory Exam Author** | 9-1-15<br>**Date** |
| **APPROVED BY:** | Pd Williams<br>**Operations Training Manager** | 10/7/15<br>**Date** |
| **APPROVED BY:** | MG<br>**Facility Representative** | 10/9/15<br>**Date** |

## SCAN OF SIGNED SCENARIO COVER SHEET

## I. OBJECTIVES

A. Given a failure of a power range detector, take corrective action for a power range instrument failure IAW AB.NIS-0001.

B. Given the order or indications of a nuclear instrument failure, DIRECT the response to the malfunction IAW S1/S2.OP-AB.NIS-0001

C. Given the indications of a reactor coolant system (RCS) malfunction or leak, perform actions as the nuclear control operator to RESPOND to the malfunctioning accordance with S1/S2.OP-AB.RC-0001.

D. Given the indications of a reactor coolant system (RCS) malfunction or leak, DIRECT the response to the malfunction in accordance with S1/S2.OP-AB.RC-0001.

E. Given the order or indications of a loss of coolant accident (LOCA), complete actions as the nuclear control operator to PERFORM the immediate response to the LOCA in accordance with the approved station procedures.

F. Given indication of a loss of coolant accident (LOCA), DIRECT the immediate response to the LOCA in accordance with the approved station procedures.

G. Given the order or a loss of coolant accident (LOCA) and plant conditions to support cold leg recirculation, perform actions as the nuclear control operator to TRANSFER to cold leg recirculation in accordance with the approved station procedures.

H. Given a loss of coolant accident (LOCA) and plant conditions to support cold leg recirculation, DIRECT actions to transfer to cold leg recirculation in accordance with the approved station procedures.

## II. MAJOR EVENTS

A. Power Ascension

B. PRNI failure

C. 20 gpm RCS leak

D. LBLOCA with failure of RHR pumps to start

E. Transfer to Cold Leg Recirc

F. Charging pump cavitation while isolating RWST

## III. SCENARIO SUMMARY

A. The crew will take the watch with the unit at 68.5% power, BOL. A power ascension at 10% per hour from 45% was placed on hold for shift turnover. Power was reduced 2 days ago for 21 SGFP control problem. Troubleshooting identified and corrected the problem, and 21 SGFP has been returned to service. The crew is directed to raise power at 10% per hour to 90% IAW

S2.OP-IO.ZZ-0004, Power Operation. Rx Engineering is bringing updated reactivity plan to control room and crew should determine its own reactivity plan. 25 CFCU is C/T for bearing replacement. 23 Condensate pump is O/S. Xenon is burning out at 60 pcm per hour.

B. The crew will continue the power ascension @ 10% / hr IAW S2.OP-IO.ZZ-0004, Power Operation.

C. After power movement has been evaluated, Power Range NI Channel IV fails high. Control rods will step in if in automatic at 72 spm. The crew will determine a load rejection is not in progress and place rods in manual. The CRS will enter S2.OP-AB.ROD-0003, Continuous Rod Motion. The crew will remove the failed channel from service. The CRS will identify applicable Tech Specs.

D. After the PRNI failure has been addressed, a 20 gpm RCS leak will occur. The crew will diagnose the leak with control console indications and alarms. The crew will place a centrifugal charging pump in service and a 45 gpm orifice in service and estimate the leak rate. The CRS will identify applicable Tech Specs.

E. After the RCS leak has been addressed, an RCS loop will catastrophically fail. RCS pressure will rapidly lower to containment pressure. Automatic Safety Injection actuation fails, and the crew will manually initiate SI. Neither RHR pump will start on the SEC signal, and the crew will block and reset 2A and 2B SECs and start both RHR pumps (CT#1). Following the reset of 2C SEC, 2C 4KV vital bus will experience an UV condition, and load in Blackout Mode. The crew will start any required ECCS equipment.

F. When RWST level reaches 15.2', the crew will transition to EOP-LOCA-3, Transfer to Cold Leg Recirculation and perform the alignment to Cold Leg Recirc. (CT#2) When isolating the RWST, both charging pumps will cavitate. The crew will secure the cavitating charging pumps.

G. The scenario will be terminated after the cavitiating charging pumps are secured, or if not identified and secured, when a charging system piping failure occurs in the Aux Building and a LOCA outside containment occurs.

## IV. INITIAL CONDITIONS

____ IC-231

## PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

| *Initial* | Description |
|---|---|
| ____ 1 | VC1and VC4 C/T |
| ____ 2 | RCPs (SELF CHECK) |
| ____ 3 | RTBs (SELF CHECK) |
| ____ 4 | MS167s (SELF CHECK) |
| ____ 5 | 500 KV SWYD (SELF CHECK) |
| ____ 6 | SGFP Trip (SELF CHECK) |
| ____ 7 | 23 CV PP (SELF CHECK) |
| ____ 8 | **25 CFCU C/T** |
| ____ 9 | **23 Condensate Pump O/S (available)** |
| ___ 10 | **IOP-4 complete up to Step 5.1.19** |
| ___ 11 | Complete Attachment 2 “Simulator Ready-for-Training/Examination Checklist.” |

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

## EVENT TRIGGERS:

| Initial | ET # | Description | |
|---|---|---|---|
| | 3 | EVENT ACTION: | kb116lck // 2SJ1 RWST TO CHG PUMP CLOSE |
| | | COMMAND: | |
| | | PURPOSE: | <update as needed> |

## MALFUNCTIONS:

| SELF-CHECK | Description | Delay Time | Initial Value | Ramp Time | Trigger | Severity |
|---|---|---|---|---|---|---|
| _____ 01 | NI0193D PR CH N44 FAILS HI/LO | N/A | N/A | N/A | RT-1 | 200 |
| _____ 02 | RC0002 RCS LEAK INTO CONTAINMENT ( equiv to 0-4 inches ) | N/A | N/A | N/A | RT-3 | 20 |
| _____ 03 | RC0001B RCS RUPTURE OF RC LOOP 22 | N/A | N/A | N/A | RT-5 | |
| _____ 04 | RP0108 FAILURE OF AUTOMATIC SI | N/A | N/A | N/A | N/A | |
| _____ 05 | RP318A2 RHR PUMP 22 Fails to Start on SEC | N/A | N/A | N/A | N/A | |
| _____ 06 | RP318A1 RHR PUMP 21 Fails to Start on SEC | N/A | N/A | N/A | N/A | |
| _____ 07 | CV0043 CHARGING LINE LEAK IN AUX BLDG | N/A | N/A | N/A | RT-9 | 550 |
| _____ 08 | CV0208A 21 CHARGING PUMP TRIP | 00:02:00 | N/A | N/A | RT-9 | |
| _____ 09 | CV0208B 22 CHARGING PUMP TRIP | 00:02:00 | N/A | N/A | RT-9 | |

## REMOTES:

| SELF-CHECK | Description | Delay Time | Initial Value | Ramp Time | Trigger | Condition |
|---|---|---|---|---|---|---|
| _____ 01 | CT195-2D 25 CFCU BKR #2 High Speed 125VDC | N/A | N/A | N/A | N/A | OFF |
| _____ 02 | CT195-3D 25 CFCU BKR #3 Low Speed 125VDC | N/A | N/A | N/A | N/A | OFF |
| _____ 03 | CV62B 22 CHG PUMP SUCTION VALVE 2CV49 | N/A | N/A | N/A | ET-3 | .00005 |
| _____ 04 | CV62A 21 CHG PUMP SUCTION VALVE 2CV44 | 00:00:02 | N/A | N/A | ET-3 | .00005 |
| _____ 05 | CT195-1D 25 CFCU BKR #1 High Speed 125VDC | N/A | N/A | N/A | N/A | OFF |
| _____ 06 | SP01A1 Prevailing Wind Speed | N/A | N/A | N/A | N/A | 10 |
| _____ 07 | SP01A2 Prevailing Wind Direction | N/A | N/A | N/A | N/A | 160 |

## OVERRIDES:

| SELF-CHECK | Description | Delay Time | Initial Value | Ramp Time | Trigger | Condition/ Severity |
|---|---|---|---|---|---|---|
| _____ 01 | C812 F DI 24CSD 2C VITAL BUS FEEDER-OPEN | N/A | N/A | N/A | N/A | ON |

| | | | | | | |
|---|---|---|---|---|---|---|
| ______ 02 | C809 F DI 23CSD 2C VITAL BUS FEEDER-OPEN | N/A | N/A | N/A | RT-7 | ON |

## OTHER CONDITIONS:

| | Description |
|---|---|
| ____ 1. | |

## V. SEQUENCE OF EVENTS

A. State shift job assignments.

B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).

C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".

D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | | | |
| **1. Power Ascension** | | | |
| | | | |
| | CRS briefs RO and PO on power ascension. | | |
| | | | |
| | RO provides reactivity plan for dilution and rod movement. | | |
| | | | |
| **Note**: Manual rod control may be used, but RCS temp will be above program and AFD will be high. | | | |
| | | | |
| | RO initiates dilution if required. | | |
| | | | |
| Note: Terr is + 0.5 °F | | | |
| | | | |
| | PO initiates a power ascension at 10% per hour. | | |
| | | | |
| | RO/PO monitor plant response to ensure power ascension is progressing as anticipated. | | |
| | | | |
| | RO either announces expected and actual auto rod movement, or withdraws rods in manual with CRS concurrence to maintain Tavg on program. | | |
| | | | |
| Proceed to next event on direction from Lead Evaluator. | | | |
| | | | |
| **2. Power Range Nuclear Instrument CH IV fails high** | | | |
| | | | |
| **Simulator Operator:** Insert **RT-1** | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| at Lead Evaluators direction.<br><br>**MALF: NI0193D, PR CH N44 fails hi**<br>**Final Val = 200** | | | |
| | | | |
| | RO reports control rods stepping in at 72 spm with no load reject in progress, places control rods in Manual (if in automatic) with concurrence from the CRS. | | |
| | | | |
| **Note:** Crew may enter procedure S2.OP-AB.ROD-0003 first, (if rods were in auto), but will transition to S2.OP-AB.NIS-0001. | | | |
| | | | |
| | CRS enters S2.OP-AB.NIS-0001, Nuclear Instrumentation Malfunction. | | |
| | | | |
| | CRS either places load change on hold or directs manual rod control. | | |
| | | | |
| | RO identifies that Power Range NI Channel IV 2N44 failed high. | | |
| | | | |
| | CRS notifies I&C of failed channel and requests assistance. | | |
| | | | |
| | PO reviews OHAs in alarm and reports they are consistent with the NI channel failure. | | |
| | | | |
| | CRS directs PO to remove 2N44 from service IAW S2.OP-SO.RPS-0001, Nuclear Instrumentation Channel Trip / Restoration. | | |
| | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | RO reports outward rod movement is unavailable until the Overpower Rod Stop is defeated, and maintains Tavg within band specified by CRS. | | |
| | | | |
| **Note:** Xenon is burning out to help temperature control. | | | |
| | | | |
| | CRS enters TSAS 3.3.1.1, Action 2 and 6. | | |
| | | | |
| | CRS directs performance of QPTR and requests Rx Engineer to perform flux map. | | |
| | | | |
| | PO checks that tripping of associated bistable(s) will NOT result in an ESF or RPS actuation. | | |
| | | | |
| | RO ensures 2N44 Channel is NOT selected on NIS Recorder 2NR45. | | |
| | | | |
| | PO places DETECTOR CURRENT COMPARATOR, UPPER SECTION, switch in PRN44 position AND ensures the following:<br>• CHANNEL DEFEAT lamp illuminates.<br>• OHA E-38, UPPER SECT DEV ABV 50% PWR, clears. | | |
| | | | |
| | PO places DETECTOR CURRENT COMPARATOR, LOWER SECTION, switch in PRN44 position and ensures the following:<br>• CHANNEL DEFEAT lamp illuminates.<br>• OHA E-46, LOWER SECT DEV ABV 50% PWR, clears. | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | | | |
| | PO places POWER MISMATCH BYPASS switch in BYPASS PR N44. | | |
| | | | |
| | PO places ROD STOP BYPASS switch in BYPASS PR N44 and ensures the following:<br>• 2RP4 - OVER POWER ROD STOP MANUAL BYPASS, CH II is illuminated.<br>• OHA E-31, PR OVERPWR ROD STOP, is clear. | | |
| | | | |
| | PO places COMPARATOR CHANNEL DEFEAT switch in N44 and ensures the following:<br>• COMPARATOR DEFEAT lamp is illuminated.<br>• OHA E-39, PR CH DEV, is clear. | | |
| | | | |
| | PO reports that remainder of procedure requires I&C support to complete. | | |
| | | | |
| | CRS contacts I&C for support if not done previously. | | |
| | | | |
| | CRS directs restoration of RCS Tavg if low out of band. | | |
| | | | |
| Continue to next even on direction from Lead Evaluator. | | | |
| | | | |
| 3. 20 gpm RCS leak | | | |
| | | | |
| **Simulator Operator**: Insert **RT-3** on direction from Lead Evaluator | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| MALF: RC0002 – RCS Leak into Containment.<br>Final Value: 20 gpm | | | |
| | | | |
| | RO reports that charging flow is rising and PZR level is lowering slowly. | | |
| | | | |
| | Crew reports reading on 2R11A containment radiation monitor is rising. | | |
| | | | |
| | RO reports unexpected OHA C-2 CNTMT SUMP PMP START. | | |
| | | | |
| | CRS enters S2.OP-AB.RC-0001, Reactor Coolant System Leak. | | |
| | | | |
| | CRS directs implementation of CAS. | | |
| | | | |
| | RO reports PZR level is lowering with maximum PDP flow. | | |
| | | | |
| | CRS enters S2.OP-AB.RAD-0001, Abnormal Radiation after OHA A-6 unexpected annunciation. | | |
| | | | |
| | RO swaps to a centrifugal charging pump IAW Step 3.14, and raises charging flow to stabilize PZR level. | | |
| | | | |
| | PO swaps to the 45 gpm letdown orifice. | | |
| | | | |
| | RO estimates leak rate. | | |
| | | | |
| | CRS initiates S2.OP-ST.RC-0008, Reactor | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | Coolant System Water Inventory Balance. | | |
| | | | |
| | CRS evaluates containment conditions and may contact Rad Pro for recommendation on CFCU operation. | | |
| | | | |
| **Role Play:** If contacted state Rad Pro recommends placing 2 CFCU's in Low Speed and 2 CFCU's in High Speed. | | | |
| | | | |
| | PO places 2 CFCU's in Low Speed and 2 CFCU's in High Speed. | | |
| | | | |
| | CRS initiates actions to locate and isolate the leak IAW Attachment 2. | | |
| | | | |
| | CRS enters TSAS 3.4.7.2.b. | | |
| | | | |
| Proceed to next event on direction from Lead Evaluator. | | | |
| | | | |
| **4. Large Break LOCA** | | | |
| | | | |
| **Simulator Operator: Delete MALF RC0002, then insert RT-5 on direction from Lead Evaluator.**<br><br>**MALF: RC0001B RCS RUPTURE OF RC LOOP 22** | | | |
| | | | |
| | RO reports indications of RCS failure (rapidly lowering RCS pressure and PZR level) | | |
| | | | |
| | RO reports the reactor has tripped or initiates | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | a manual reactor trip. | | |
| | | | |
| **5. Auto SI fails to actuate** | | | |
| | | | |
| | RO performs Immediate Actions of TRIP-1<br>- Confirms the Rx trip<br>- Trips the Main Turbine<br>- Reports at least one 4KV Vital bus energized.<br>Reports a demand for SI exists, but SI has not initiated, and manually initiates SI. | | |
| | | | |
| | CRS and RO verify performance of immediate actions. | | |
| | | | |
| | PO reports all AFW pumps running, and requests permission to throttle AFW flow while maintaining 22E4 lbm/hr or 9% level in at least one SG NR level. | | |
| | | | |
| **6. 21 and 22 RHR pumps fail to start on SEC** | | | |
| | | | |
| | PO performs SEC Loading Verification along with safeguards valve alignment, and reports neither RHR pump started. | | |
| | | | |
| | RO blocks 2A and 2B SECs. | | |
| | | | |
| | PO resets 2A and 2B SECs. | | |
| | | | |
| | RO starts 21 and 22 RHR pumps. | | |
| | | | |
| **CT#1 (CT-5) Manually start at least one low-head ECCS pump** | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| before transition out of TRIP-1.<br><br>SAT_____ UNSAT_____ | | | |
| | | | |
| | PO reports 21 and 22 AFW pumps running. | | |
| | | | |
| | PO reports valve groups in Table B are in safeguards position. | | |
| | | | |
| | RO reports 21 and 22CA330s are shut. | | |
| | | | |
| | RO reports containment pressure has not remained below 15 psig. | | |
| | | | |
| | RO backs up Phase B and Spray initiation by turning both keys. | | |
| | | | |
| | RO reports both containment spray pumps running and MSLI has occurred. | | |
| | | | |
| | RO stops all RCPs if not previously performed per CAS. | | |
| | | | |
| | PO reports all valves in Table D in safeguards position. | | |
| | | | |
| | PO reports no high steam flow condition. | | |
| | | | |
| | CRS directs implementation of the ECG. | | |
| | | | |
| | PO reports all 4KV vital buses energized. | | |
| | | | |
| | RO reports CAV in AP mode. | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | | | |
| | RO runs 2 switchgear supply and one exhaust fan. | | |
| | | | |
| | RO reports 2 or more CCW pumps running. | | |
| | | | |
| | RO reports RCS not aligned for Cold Leg Recirc. | | |
| | | | |
| | RO reports charging flow is at least 100 gpm on SI systems charging flow meter. | | |
| | | | |
| | RO reports RCS pressure is <1660 psig (adverse), and SI flow is at least 100 gpm on 21 or 22 SI pump flow meter. | | |
| | | | |
| | RO reports RCS pressure < 420 psig (adverse) and RHR flow is at least 300 gpm on both 21 and 22 SJ49s. | | |
| | | | |
| | RO closes charging pump mini flow valves with RCS pressure <1500 psig if not previously performed per CAS. | | |
| | | | |
| | PO reports total AFW flow is 22E4 lbm/hr or 15% level in at least one SG NR level, and maintains 22E4 lbm/hr until at least one SG NR level is >15% (adverse), then maintains 15-33% level. | | |
| | | | |
| **Note:** SG pressures will drop over scenario due to LBLOCA, and AFW will rise with no action from PO. | | | |
| | | | |
| | RO reports no RCPs running, and RCS | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | temperature is not stable at or trending to 547°F. | | |
| | | | |
| | PO reports no steam dumping in progress. | | |
| | | | |
| | CRS determines MSLI has been performed. | | |
| | | | |
| | RO reports both RTBs are open. | | |
| | | | |
| | RO reports both PZR PORVs are shut with their Block Valves open. | | |
| | | | |
| | PO reports no indication of faulted SGs. | | |
| | | | |
| **Note:** SG pressures will drop over scenario due to LBLOCA. | | | |
| | | | |
| | PO/RO report no indications of a steam generator tube rupture. | | |
| | | | |
| | RO reports 2 or more channels in Table F are in warning or alarm. | | |
| | | | |
| | CRS transitions to EOP-LOCA-1, Loss of Reactor Coolant. | | |
| | | | |
| **Note:** CFST monitoring will commence after the transition out of TRIP-1. STA will report to the control room 10 minutes after being paged. When the FRTS RED path is identified, the CRS goes to FRTS-1. When the FRCE PURPLE path is identified, the | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| CRS will go to FRCE-1. | | | |
| | | | |
| | STA reports CFST status | | |
| | | | |
| **FRTS steps** | | | |
| | | | |
| | CRS enters 2-EOP-FRTS-1, Response to Imminent Pressurized Thermal Shock Conditions. | | |
| | | | |
| | RO reports RCS pressure is < 420 psig. | | |
| | | | |
| | CRS returns to procedure in effect. | | |
| | | | |
| **FRCE steps** | | | |
| | | | |
| | CRS enters 2-EOP-FRCE-1, Response to Excessive Containment Pressure. | | |
| | | | |
| | PO verifies Phase A valves shut. | | |
| | | | |
| | RO verifies 2VC5 and 2VC6 shut, containment pressure is > 15 psig, and LOCA-5 not in effect. | | |
| | | | |
| | RO reports containment spray and Phase B status, and all RCPs are stopped. | | |
| | | | |
| | RO reports CFCU status. | | |
| | | | |
| | PO reports all MSIVs are shut. | | |
| | | | |
| | PO reports no faulted SGs | | |
| | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | CRS returns to procedure in effect. | | |
| | | | |
| **LOCA-1 steps** | | | |
| | | | |
| | Crew re-verifies no faulted or ruptured SGs exist. | | |
| | | | |
| | RO resets SI, Phase A, and Phase B isolations. | | |
| | | | |
| | RO opens 21 and 22CA330s. | | |
| | | | |
| | PO resets each SEC. | | |
| | | | |
| **7. 2C 4KV vital bus UV** | | | |
| | | | |
| **Note:** Event 7 may not be inserted if crew does not reach SEC reset steps in LOCA-1. | | | |
| | | | |
| **Simulator Operator**: Insert **RT-7** after all SECs have been reset. This will deenergize 2C 4KV vital bus. The 2C EDG will start and load the bus in Blackout mode. | | | |
| | | | |
| | PO reports UV condition on 2C 4KV vital bus, and EDG is loading Blackout loads. | | |
| | | | |
| | Crew performs actions of Table B for 2C SEC:<br>PO verifies loading complete<br>PO resets 2B SEC<br>CRS directs starting of safeguards loads. | | |
| | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | PO resets 230V control centers. | | |
| | | | |
| | RO resets SGBD Sample Isolation Bypass and opens 21-24SS94s. | | |
| | | | |
| | CRS directs chemistry to sample SGs for activity and boron. | | |
| | | | |
| | Crew verifies proper PORV status and that SI flow cannot be reduced. | | |
| | | | |
| | Crew verifies CS flow is indicated. | | |
| | | | |
| | RO verifies RHR pumps running for LBLOCA. | | |
| | | | |
| | Crew re-verifies no faulted SGs. | | |
| | | | |
| | CRS evaluates continued EDG running status. | | |
| | | | |
| | CRS transitions to 2-EOP-LOCA-3, Transfer to Cold Leg Recirculation when RO announces 2/4 RWST level have reached 15.2 ft. | | |
| | | | |
| **Mark time RWST lo level alarm is received for critical task performance:**<br><br>_____:_____:_____ | | | |
| | | | |
| 4. Transfer to Cold Leg Recirculation | | | |
| | | | |
| | RO reports Containment Sump Level Lights illuminated >62%. | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | | | |
| | RO depresses SUMP AUTO ARMED PB for 21/22SJ44s. | | |
| | | | |
| | PO removes lockouts from 2SJ67, 2SJ68, 2SJ69. | | |
| | | | |
| | RO reports 21 and 22SJ44 open. | | |
| | | | |
| | RO reports both RHR pumps running. | | |
| | | | |
| | RO shuts 2SJ69. | | |
| | | | |
| **Mark time 2SJ69 close PB depressed.** _____:_____:_____ | | | |
| | | | |
| **CT#2 (CT-36) Transfer to Cold Leg Recirculation such that at least one train of ECCS is in operation in the recirculation mode within the following time frames.**<br>**1. From RWST lo level alarm to initiating closed on 2SJ69 - ≤3.7 minutes**<br>**2. From RWST lo level alarm to one containment spray pump stop ≤5.5 minutes**<br>**3. From RWST lo level alarm to switchover completion (includes restarting ECCS pumps if stopped on RWST lo-lo level) ≤ 11.2 minutes.** | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| SAT _____ UNSAT _____ | | | |
| | | | |
| | RO verifies or performs SI reset actions. | | |
| | | | |
| | RO resets SECs and 230V MCCs. | | |
| | | | |
| | RO stops 22 Containment Spray Pump. | | |
| | | | |
| **Mark time of 22 CS pump stop.** <br> _____:_____:______ <br> **< 5.5 mins after RWST Lo Level CT#2(2)** | | | |
| | | | |
| | RO closes 21/22RH19s. | | |
| | | | |
| | RO stops 23 charging pump. | | |
| | | | |
| | CRS goes to step 11 based on all Vital Buses energized. | | |
| | | | |
| | CRS goes to step 28 if any EDG is supplying its vital bus. | | |
| | | | |
| Step 28 | RO reports 3 SW pumps running. | | |
| | | | |
| Step 28 | PO reports both CCHXs are in service . | | |
| | | | |
| Step 28.2 | RO reports 23 CCW pump is NOT in service. | | |
| | | | |
| Step 28.3 | RO reports 22 CS pump is NOT in service | | |
| | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| Step 28.3 | RO starts 23 CCW pump. | | |
| Remaining recirc alignment steps are common to steps 28/29 if an EDG is supplying its vital bus, or steps 11-14 if all 4KV vital buses energized from off site power. | RO reports 21CC16 and 22CC16 are open. | | |
| | RO shuts 2SJ67 and 2SJ68. | | |
| | RO reports 2RH1 and 2RH2 are shut. | | |
| | RO reports 22 RHR pump is running and opens 22SJ45. | | |
| | RO reports 21 RHR pump in service and opens 21SJ45. | | |
| | RO reports 21SJ113 and 22SJ113. | | |
| | Reports 21 and 22 Safety Injection Pumps and 21 and 22 Charging Pump are running. | | |
| **Mark time ECCS pumps verified in operation.**<br>_____:_____:______<br>**<11.7 mins after RWST Lo Level CT#2(3)** | | | |
| | PO removes Lockout from 2SJ30. | | |
| | RO closes 2SJ30, 2SJ1, and 2SJ2 | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| **Simulator Operator: Simulator Operator:** Ensure **ET-3** is TRUE when the close PB for 2SJ1 is depressed. This inserts the charging pump cavitation. Pump amp oscillations between 48-90 amps will occur, and BIT flow will lower to 0. | | | |
| | | | |
| | RO reports indications of charging pump cavitation of fluctuating charging pump amps, seal injection flow alarms, and loss of BIT flow. | | |
| | | | |
| | RO stops 21 and 22 charging pumps. | | |
| | | | |
| **CT#3 Trip the cavitating charging pump prior to pump/system piping damage.**<br><br>**SAT _____ UNSAT _____** | | | |
| | | | |
| **Simulator Operator:** IF the crew does **NOT** trip 22 charging pump within 3 minutes of the initiation of cavitation, **THEN** insert **RT-9.** This simulates a failure of the charging system piping in the Aux Building resulting in a LOCA outside containment, then trips 22 charging pump after an additional 2 minute delay.<br>**MALF: CV0043 Charging Line Leak in Aux Bldg**<br>**Final Value: 550**<br>**MALFS:** | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| **CV0208A 21 Chg Pump trip**<br>**CV0208B 22 Chg Pump trip**<br>**Delay: 2 minutes**<br>**Final Value: True** | | | |
| | | | |
| **TERMINATE** the scenario when both charging pumps have been secured, **OR** if the pumps are not secured, after crew recognizes LOCA in Aux bldg, **OR** at Lead Evaluators discretion. | | | |
| | | | |
| **Note**: Some of the following steps may be performed depending on when, or if, the crew recognizes the charging pump cavitation, and should be evaluated accordingly. | | | |
| | | | |
| | RO places 21 and 22RH29 controllers in Manual and ensures valves are shut. | | |
| | | | |
| | CRS implements FRPs as necessary. | | |
| | | | |
| | CRS dispatches operator to tag SJ44 breakers open. | | |
| | | | |
| | RO verifies Phase A and Phase B are reset, and CA330s are open. | | |
| | | | |

## VI. SCENARIO REFERENCES

A. Alarm Response Procedures (Various)

B. Technical Specifications

C. Emergency Plan (ECG)

D. OP-AA-101-111-1003, Use of Procedures

E. S2.OP-IO.ZZ-0004, Power Operation

F. S2.OP-AB.ROD-0003, Continuous Rod Motion

G. S2.OP-AB.NIS-0001, Nuclear Instrumentation System Malfunction

H. S2.OP-AB.RC-0001, Reactor Coolant System Leak

I. 2-EOP-TRIP-1, Reactor Trip or Safety Injection

J. 2-EOP-LOCA-1, Loss of Reactor Coolant

K. 2-EOP-LOCA-3, Transfer to Cold Leg Recirculation

L. 2-EOP-FRTS-1, Response to Imminent Pressurized Thermal Shock Condition

M. 2-EOP-FRCE-1, Response to High Containment Pressure

**ATTACHMENT 1**
**UNIT TWO PLANT STATUS**
**TODAY**

MODE: 1 POWER: 67.5% RCS BORON: 1235 MWe 780

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Control Bank D 168 steps withdrawn
Xenon burning out at 60 pcm per hour
Power was reduced to 45% 2 days ago for SGFP problem, which was fixed.

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

3.6.2.3 action a (25 CFCU) expires in 166 hours.

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

Power ascension @ 10% per hour IAW S2.OP-IO.ZZ-0004

ABNORMAL PLANT CONFIGURATIONS:

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.
No penalty minutes in the last 24 hrs.
25 CFCU C/T for motor bearing replacement.

PRIMARY:

None

SECONDARY:

23 Condensate pump O/S

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

None

## ATTACHMENT 2

## SIMULATOR READY FOR TRAINING CHECKLIST

____ 1. Verify simulator is in "TRAIN" Load

____ 2. Simulator is in RUN

____ 3. Overhead Annunciator Horns ON

____ 4. All required computer terminals in operation

____ 5. Simulator clocks synchronized

____ 6. All tagged equipment properly secured and documented

____ 7. TSAS Status Board up-to-date

____ 8. Shift manning sheet available

____ 9. Procedures in progress open and signed-off to proper step

____ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced

____ 11. Required chart recorders advanced and ON (proper paper installed)

____ 12. All printers have adequate paper AND functional ribbon

13. Required procedures clean

____ 14. Multiple color procedure pens available

____ 15. Required keys available

____ 16. Simulator cleared of unauthorized material/personnel

____ 17. All charts advanced to clean traces and chart recorders are on.

____ 18. Rod step counters correct (channel check) and reset as necessary

____ 19. Exam security set for simulator

____ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter

____ with Baseline Data filled out

____ 21. Shift logs available if required

____ 22. Recording Media available (if applicable)

____ 23. Ensure ECG classification is correct

____ 24. Reference verification performed with required documents available

____ 25. Verify phones disconnected from plant after drill.

____ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.

____ 27. Ensure sufficient copies of ECG paperwork are available.

## ATTACHMENT 3

## CRITICAL TASK METHODOLOGY

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.
The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...

- degradation of any barrier to fission product release
- degraded emergency core cooling system (ECCS) or emergency power capacity
- a violation of a safety limit
- a violation of the facility license condition
- incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
- a significant reduction of safety margin beyond that irreparably introduced by the scenario

II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...

- effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
- recognize a failure or an incorrect automatic actuation of an ESF system or component.
- take one or more actions that would prevent a challenge to plant safety.
- prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation.

## ATTACHMENT 4

## SIMULATOR SCENARIO REVIEW CHECKLIST

**SCENARIO IDENTIFIER: 14-01 ILOT NRC ESG-1 REVIEWER: B Blose**

| Initials | | Qualitative Attributes |
|---|---|---|
| BB | 1. | The scenario has clearly stated objectives in the scenario. |
| BB | 2. | The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events. |
| BB | 3. | The scenario consists mostly of related events. |
| BB | 4. | Each event description consists of:<br>• the point in the scenario when it is to be initiated<br>• the malfunction(s) that are entered to initiate the event<br>• the symptoms/cues that will be visible to the crew<br>• the expected operator actions (by shift position)<br>• the event termination point |
| BB | 5. | No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event. |
| BB | 6. | The events are valid with regard to physics and thermodynamics. |
| BB | 7. | Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives. |
| BB | 8. | The simulator modeling is not altered. |
| BB | 9. | All crew competencies can be evaluated. |
| BB | 10. | The scenario has been validated. |
| BB | 11. | If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario. |
| BB | 12. | ESG-PSA Evaluation Form is completed for the scenario at the applicable facility. |

## ATTACHMENT 4

## SIMULATOR SCENARIO REVIEW CHECKLIST

| Initial | | Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d) |
|---|---|---|
| GG | 3 | Malfunctions after EOP entry: 1-2 |
| GG | 2 | Abnormal Events: 2-4 |
| GG | 1 | Major Transients: 1-2 |
| GG | 2 | EOPs entered/requiring substantive actions: 1-2 |
| GG | 2 | EOP contingencies requiring substantive actions: 0-2 |
| GG | 3 | EOP based Critical tasks: 2-3 |

**COMMENTS:**

**ATTACHMENT 5**

**ESG CRITICAL TASKS**

**14-01 ILOT NRC ESG-1**

**CT#1 (CT-5) Manually start at least one low-head ECCS pump before transition out of TRIP-1.**

Bases: Failure to manually start at least one low-head ECCS pump under the postulated conditions constitutes misoperation or incorrect crew performance in which the crew does not prevent "degraded emergency core cooling system (ECCS) ...capacity." In this case, at least one low-head ECCS pump can be manually started from the control room. Therefore, failure to manually start a low-head ECCS pump also represents a failure by the crew to "demonstrate the following abilities:

- Effectively direct or manipulate engineered safety feature (ESF) controls that would prevent a significant reduction of safety margin (beyond that irreparably introduced by the scenario)
- Recognize a failure or an incorrect automatic actuation of an ESF system or component"

**CT#2 (CT-36) Transfer to Cold Leg Recirculation such that at least one train of ECCS is in operation in the recirculation mode within the following time frames.**
**1. From RWST lo level alarm to initiating closed on 2SJ69 - ≤3.7 minutes**
**2. From RWST lo level alarm to one containment spray pump stop ≤5.5 minutes**
**3. From RWST lo level alarm to switchover completion (includes restarting ECCS pumps if stopped on RWST lo-lo level) ≤ 11.2 minutes.**

**Bases:** Omission or incorrect performance of this task results in "direct adverse consequences or significant degradation in the mitigative capability of the plant." Failure to transfer to cold leg recirculation before the RWST inventory is totally depleted results in the loss of all pumped safety injection and containment spray when the RWST empties. Provided that transfer to cold leg recirculation is possible (as is postulated in the plant conditions), a failure to transfer resulting in loss of pumped injection and containment constitutes misoperation or incorrect crew performance which fails to prevent "degraded ECCS...capacity." It also constitutes "a significant reduction of safety margin beyond that irreparably introduced by the scenario. "Failure to transfer to cold leg recirculation under the postulated plant conditions can result in unnecessary challenges to the following CSFs:

- Core cooling · Containment

Thus, failure to transfer represents a demonstrated inability by the crew to "take one or more actions that would prevent a challenge to plant safety."

**CT#3 Trip the cavitating charging pump prior to pump/system piping damage.**

Bases: Failure to trip a cavitating charging pump during the transfer to cold leg recirculation when indications of cavitation are present, leads to the possibility of system damage and the advent of a loss of coolant outside the containment building. (As is the case in this scenario.)

**EVENTS LEADING TO CORE DAMAGE**

| Y/N | Event | Y/N | Event |
|---|---|---|---|
| N | TRANSIENTS with PCS Unavailable | N | Loss of Service Water |
| N | Steam Generator Tube Rupture | N | Loss of CCW |
| N | Loss of Offsite Power | N | Loss of Control Air |
| N | Loss of Switchgear and Pen Area Ventilation | N | Station Black Out |
| Y | LOCA | | |

**COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY**

| Y/N | COMPONENT, SYSTEM, OR TRAIN | Y/N | COMPONENT, SYSTEM, OR TRAIN |
|---|---|---|---|
| N | Containment Sump Strainers | N | Gas Turbine |
| N | SSWS Valves to Turbine Generator Area | N | Any Diesel Generator |
| N | RHR Suction Line valves from Hot Leg | N | Auxiliary Feed Pump |
| N | CVCS Letdown line Control and Isolation Valves | N | SBO Air Compressor |

**OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE**

| Y/N | OPERATOR ACTION |
|---|---|
| N | Restore AC power during SBO |
| N | Connect to gas turbine |
| N | Trip Reactor and RCPs after loss of component cooling system |
| Y | Re-align RHR system for re-circulation |
| N | Un-isolate the available CCW Heat Exchanger |
| N | Isolate the CVCS letdown path and transfer charging suction to RWST |
| N | Cooldown the RCS and depressurize the system |
| N | Isolate the affected Steam Generator that has the tube rupture(s) |
| N | Early depressurize the RCS |
| N | Initiate feed and bleed |

Complete this evaluation form for each ESG.